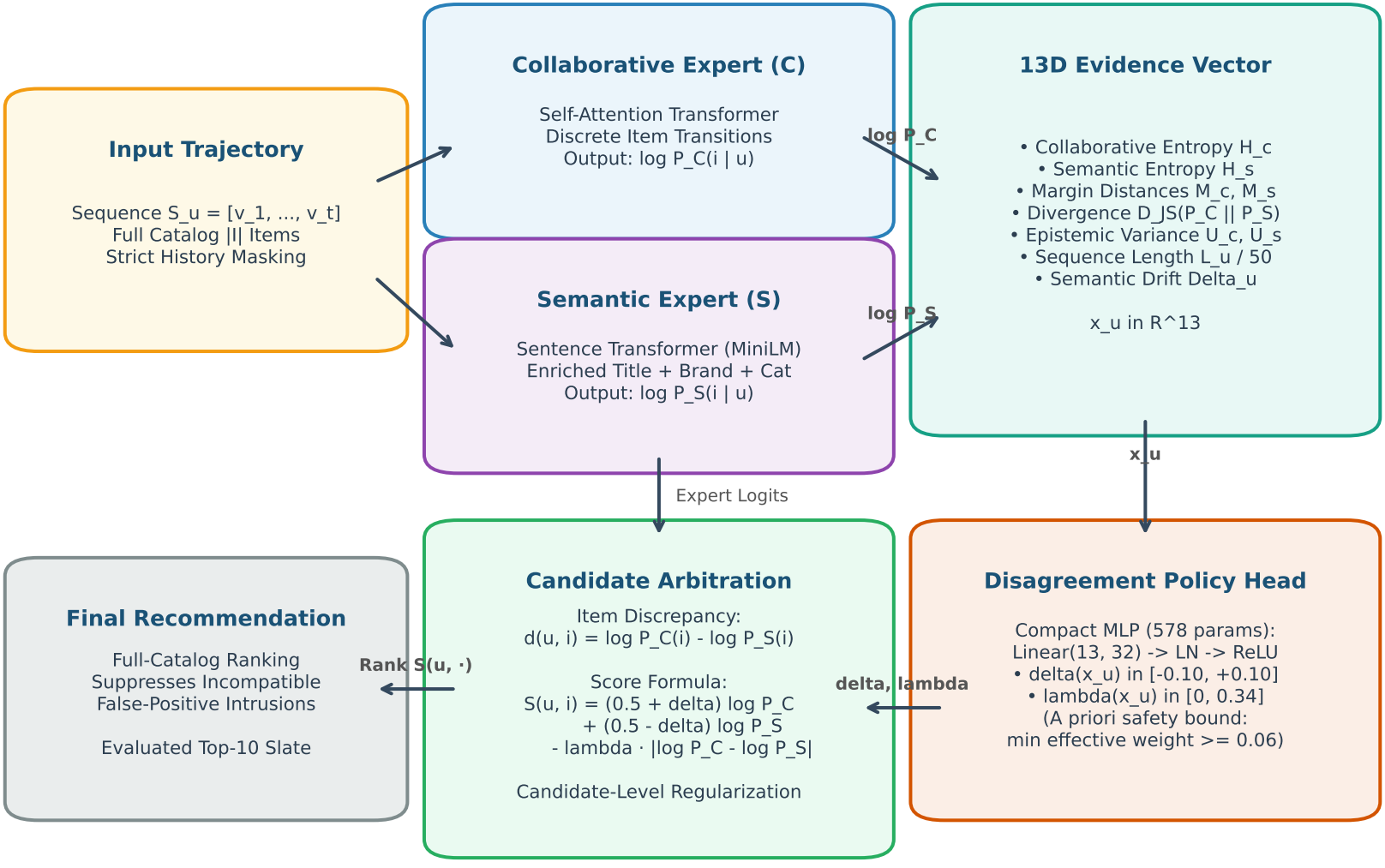
Input Trajectory
Sequence S_u = [v_1, ..., v_t]
Full Catalog |I| Items
Strict History Masking
Collaborative Expert (C)
Self-Attention Transformer
Discrete Item Transitions
Output: log P_C(i | u)
Semantic Expert (S)
Sentence Transformer (MiniLM)
Enriched Title + Brand + Cat
Output: log P_S(i | u)
log P_C
log P_S
13D Evidence Vector
• Collaborative Entropy H_c
• Semantic Entropy H_s
• Margin Distances M_c, M_s
• Divergence D_JS(P_C || P_S)
• Epistemic Variance U_c, U_s
• Sequence Length L_u / 50
• Semantic Drift Delta_u
x_u in R^13
x_u
Expert Logits
Candidate Arbitration
Item Discrepancy:
d(u, i) = log P_C(i) - log P_S(i)
Score Formula:
S(u, i) = (0.5 + delta) log P_C
+ (0.5 - delta) log P_S
- lambda · |log P_C - log P_S|
Candidate-Level Regularization
delta, lambda
Disagreement Policy Head
Compact MLP (578 params):
Linear(13, 32) -> LN -> ReLU
• delta(x_u) in [-0.10, +0.10]
• lambda(x_u) in [0, 0.34]
(A priori safety bound:
min effective weight >= 0.06)
Rank S(u, ·)
Final Recommendation
Full-Catalog Ranking
Suppresses Incompatible
False-Positive Intrusions
Evaluated Top-10 Slate